

Cross-Domain Research Report

Physics-Informed Neural Networks for volatility trading and market prediction

Cross-Domain Analysis | 56 papers | 2 bridges

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Research Overview

This research investigates the intersection of Networks / Learning / Neural and Fractional / Applications / Situated, responding to the query: "Physics-Informed Neural Networks for volatility trading and market prediction"

At first glance, these domains appear unrelated. Networks / Learning / Neural is primarily concerned with We trained a large, deep convolutional neural network to classify the 1.2 million high-resolution images in the ImageNet LSVRC-2010 contest into the 1..., while Fractional / Applications / Situated focuses on The Aion Framework presents a bold, unified dimensional hypothesis that reinterprets AI consciousness, quantum mechanics, cosmology, and human immorta....

However, our analysis of 56 papers across both domains, including 2 papers that bridge both fields, reveals several non-obvious connections.

Key bridging concepts include: evidence, cross ref, vol august, electronic journalfinancial, development, economic, vol july, electronic, april cross, financial development.

The most significant cross-domain paper is "Fractional Brownian Motions, Fractional Noises and Applications" (7678 citations), which directly addresses both domains.

The following sections detail the methodology, key findings, and implications of this cross-domain analysis.

Methodology

We employed a systematic cross-domain literature review methodology:

- Source Querying**: Papers were ingested from OpenAlex and arXiv using domain-specific search queries. A total of 56 papers were collected across both domains.

2. **Deduplication**: Papers were deduplicated by DOI and fuzzy title matching, ensuring each paper was analyzed only once.
3. **Concept Extraction**: Key concepts were extracted from titles and abstracts using term frequency analysis with bigram detection.
4. **Cross-Domain Bridging**: We identified papers that contain concepts from both domains, serving as bridges between the fields.
5. **Synthesis**: Findings were synthesized to identify non-obvious connections, shared methodologies, and potential applications across domains.

Domain Analysis

Domain A: Networks / Learning / Neural

Papers analyzed: 33

Key concepts (by frequency):

- cross ref: 666 occurrences
- cross: 339 occurrences
- electronic: 211 occurrences
- financial: 96 occurrences
- scheduling: 94 occurrences
- development: 75 occurrences
- systems: 72 occurrences
- parallel: 71 occurrences
- electronic journalthe: 68 occurrences
- economic growth: 66 occurrences

- learning
- march cross
- vol oct
- science vol
- applications
- nov cross
- systems vol
- september cross

- "ImageNet classification with deep convolutional neural networks" (75705 cites, 2017)
- "Gradient-based learning applied to document recognition" (57987 cites, 1998)
- "" (50260 cites, 2021)
- "Emergence of Scaling in Random Networks" (36303 cites, 1999)
- "Physics-informed neural networks: A deep learning framework for s" (16399 cites, 2018)

Domain B: Fractional / Applications / Situated

Papers analyzed: 23

Key concepts (by frequency):

- cross ref: 308 occurrences
- electronic: 211 occurrences
- cross: 160 occurrences
- financial: 101 occurrences
- development: 78 occurrences
- economic: 69 occurrences
- electronic journalthe: 68 occurrences
- economic growth: 66 occurrences
- fractional: 64 occurrences
- growth: 58 occurrences

- brownian motion
- ssrn
- market
- capital
- economics vol
- vol september
- science
- journalfinancial

- "Aion Framework: Dimensional Emergence of AI Consciousness, Observ" (14221 cites, 2023)
- "Fractional Brownian Motions, Fractional Noises and Applications" (7678 cites, 1968)
- "Situated Knowledges: The Science Question in Feminism and the Pri" (7010 cites, 1988)
- "The Problem of Pattern and Scale in Ecology: The Robert H. MacArt" (6743 cites, 1992)
- "Whatever next? Predictive brains, situated agents, and the future" (5824 cites, 2013)

Cross-Domain Findings

Found 2 papers that bridge both domains.

Shared Concepts

Concepts appearing in both domains: evidence, cross ref, vol august, electronic journalfinancial, development, economic, vol july, electronic, april cross, financial development, august cross, cross, financial, july cross, scholar

Domain Scale and Scope [confidence: 0.95]

Domain A contains 33 papers with 2 distinct sources. Domain B contains 23 papers. The combined corpus spans 1968-2026.

Cross-Domain Bridge: Fractional Brownian Motions, Fractional Noises and Applicati [confidence: 0.90]

This paper (7678 citations) directly addresses both domains. Abstract: Previous article Next article Fractional Brownian Motions, Fractional Noises and ApplicationsBenoit B. Mandelbrot and John W. Van NessBenoit B. Mandelbrot and John W. Van Nesshttps://doi.org/10.1137/1010093PDFBibTexSections ToolsAdd to favoritesExport CitationTrack CitationsEmail SectionsAbout[1] I....

Cross-Domain Bridge: Scientific Machine Learning Through Physics-Informed Neural [confidence: 0.73]

This paper (2252 citations) directly addresses both domains. Abstract: Abstract Physics-Informed Neural Networks (PINN) are neural networks (NNs) that encode model equations, like Partial Differential Equations (PDE), as a component of the neural network itself. PINNs are nowadays used to solve PDEs, fractional equations, integral-differential equations, and stochastic...

Shared Conceptual Framework [confidence: 0.80]

The following concepts appear in both domains: evidence, cross ref, vol august, electronic journalfinancial, development, economic, vol july, electronic, april cross, financial development, august cross, cross, financial, july cross, scholar. This shared vocabulary suggests underlying methodological connections that could enable knowledge transfer between the fields.

Foundational Papers in Domain A [confidence: 0.90]

- ? "ImageNet classification with deep convolutional neural networks" (75705 cites, 2017)
- ? "Gradient-based learning applied to document recognition" (57987 cites, 1998)
- ? "" (50260 cites, 2021)

Foundational Papers in Domain B [confidence: 0.90]

- ? "Aion Framework: Dimensional Emergence of AI Consciousness, Observer-In" (14221 cites, 2023)
- ? "Fractional Brownian Motions, Fractional Noises and Applications" (7678 cites, 1968)
- ? "Situated Knowledges: The Science Question in Feminism and the Privileg" (7010 cites, 1988)

Identified Research Gap [confidence: 0.75]

Despite 56 papers across both domains, only 2 papers explicitly bridge them. This represents a significant research opportunity ? the methodological tools of Networks / Learning / Neural could be applied to Fractional / Applications / Situated with novel results.

Implications

- The shared concepts (evidence, cross ref, vol august, electronic journalfinancial, development) suggest that methodological transfer between these domains is feasible.
- Existing cross-domain work (2 papers) provides a foundation for deeper integration of these fields.
- The research mesh successfully identified non-obvious connections between domains that would be difficult to discover through manual literature review.

Future Research Directions

- Apply learning techniques from Domain A to Domain B problems.
- Investigate whether brownian motion methods from Domain B can enhance Domain A approaches.
- Develop unified frameworks leveraging the shared concept of evidence.
- Scale the analysis to 500+ papers per domain for more robust cross-domain detection.

Research Mesh Statistics

Metric	Value
Total Papers	154
Domain A Papers	33
Domain B Papers	23
Cross-Domain Papers	2
Graph Nodes	339
Graph Edges	20572
Vault Paper Notes	5
Vault Doctrine Notes	123
Domains Covered	4

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